The Spotted Lanternfly (*Lycorma delicatula*): A Comprehensive Review of Biology, Impact, Management, and Data Resources

I. Executive Summary

The spotted lanternfly (SLF) *Lycorma delicatula* (White, 1845) is an invasive planthopper native to Asia that has become a significant threat to U.S. agriculture and forestry since its initial detection in Pennsylvania in 2014. Since its introduction, it has spread rapidly across multiple states, causing damage to crops, ornamental plants, and native tree species.

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This comprehensive review summarizes key aspects of spotted lanternfly biology, identification, distribution, and management. It examines the species' life cycle, feeding behavior, host range, and impacts across affected industries. Ongoing research continues to evaluate the full extent of its economic and ecological damage, as well as its potential role in plant disease dynamics.

II. Introduction

The SLF is an invasive insect that has recently established in North America. Native to China, India, and Vietnam, it first spread throughout South Korea and has since expanded into several eastern states within the U.S. Feeding on over 170 plant species and the ability to reproduce rapidly and in large numbers has caused concern since its arrival. First detected in 2014, the SLF likely entered the U.S. on imported materials. Infestation has since expanded across multiple states and poses major economic and ecological risks especially grapevines.

III. Biology and Identification of the Spotted Lanternfly (*Lycorma delicatula*)

Life Cycle: Egg, Nymph, and Adult Stages

The SLF undergoes one generation per year, progressing through four main life stages: egg, early instar nymph, fourth instar nymph, and adult. Although later nymphal stages resemble adults in appearance, nymphs are not strong fliers; instead, they typically jump or glide short distances when disturbed.

Physical Description and Distinguishing Features

- **Adults:** The adult SLF is roughly 1-inch long and 1/2-inch wide with large visually striking wings.[1] Forewings are brown with black spots at the front, and a speckled band near the rear.[1] Their hindwings are a vibrant scarlet with black spots towards thefront and white/black bars near the rear.[1] The SLF has a yellow abdomen with black bars. [1]
- **Eggs:** Newly laid egg masses, each containing approximately 30–50 eggs, resemble wet, gray putty before drying to a dull brown and taking on the appearance of cracked mud.[1] These masses are typically about 1 inch in length and can be observed from fall through spring on a wide range of smooth surfaces, including both man-made objects (e.g., vehicles) and natural substrates such as tree bark..[1]
- **Nymphs:** The spotted lanternfly progresses through four nymphal instars prior to reaching adulthood. The first three instars, collectively referred to as early instar nymphs, are similar in appearance and typically hatch between March and April [6]. During these stages, nymphs are approximately 1/8 inch in length and are characterized by a black body with distinct white spots.[6] Following the third instar, the insect develops into the fourth instar nymph, usually by June [6]. This stage is larger, approximately 1/2 inch in

length, and is easily distinguished by its red coloration with white spots and contrasting black striping.[6]

Host Plants and Feeding Behavior

The tree-of-heaven (*Ailanthus altissima*) is a preferred host of the spotted lanternfly (SLF) [1]. This species is characterized by smooth bark in younger trees that appears light brown to gray, developing a more textured surface with age. Leaves can reach up to four feet in length and are pinnately compound, consisting of numerous leaflets arranged along a central rachis. Spotted lanternflies feed gregariously on the phloem of more than 172 documented host plant species worldwide across all life stages, although host preference narrows as individuals mature [10].

Geographical Distribution (Global and United States)

The native range of *Lycorma delicatula* includes China, India, and Vietnam. It was first reported in South Korea in 1932 and has since become abundant there.[2][10] The SLF was first detected in the U.S. in 2014, but was thought to have been established for 2-3 years.[4] The spotted lanternfly (*Lycorma delicatula*), first detected in Pennsylvania in 2014, has established populations in 19 states and the District of Columbia, with the heaviest infestations in the Northeast and Mid-Atlantic and continued expansion into the Midwest and South. Its spread is driven largely by human-mediated movement, as individuals readily hitchhike on vehicles, firewood, and equipment. The species is strongly associated with tree-of-heaven (*A.altissima*), which facilitates establishment and dispersal. In response, multiple states have implemented quarantines to limit the transport of materials that may harbor egg masses.Scientist continue to determine their distribution patterns within different regions in order to prevent the spread SLFs in areas where they could thrive.

Table 1: Key Characteristics of Spotted Lanternfly Life Stages

Life Stage	Size (approx.)	Color/Appearance	Key Distinguishing Features	Typical Location
Eggs	1 in long	Gray or a dull brown	In masses during October-June Resembles wet putty before dulling and mirrors cracked mud	Smooth surfaces such as trees, bricks, stones, fences, grills, and vehicles
Early Instar Nymph	1/8 in long	Black with white spots	Strong jumpers when prodded or frightened Hatch in April	Early stages feed on tree-of-heaven or new plant growth such as stems and foliage
Fourth Instar Nymph	½ in long	Red with white dots and black stripes	Strong jumpers when prodded or frightened Clustered together Closer to June	Base or branches of tree-of-heaven
Adults	1 in long ½ in wide	Brown, yellow abdomen with black bars	Striking wings they have black spots, hind wings are scarlet with black spots, and white/black bars at the rear Gather in large numbers Glides rather than fly	Trunk of trees and grapevines

IV. Disease

An important unknown regarding the SLF is its potential to transmit plant diseases. Although no evidence currently indicates that SLF acts as a disease vector in either the United States or Korea [8], it is taxonomically related to other planthopper groups, such as Delphacidae and Cixiidae, which are known to transmit plant viruses. As a result, research efforts, including studies initiated in vineyards in 2019, have focused on evaluating its capacity for virus transmission. To date, these investigations have not demonstrated any role of SLF in spreading plant pathogens.

Symptoms SLFs leave on Plants

- **Leaf Symptoms:** When the SLF feeds on plants, they excrete a sticky, sugary liquid called "honeydew." This substance leaves behind a shiny, sticky residue that attracts other insects, promotes sooty mold growth, and further harms plants, developing a vinegar-like smell[1]
- **Differentiation from Nutritional Deficiencies:** Nymphs and adults of the SLF feed on the sap of a host plant through the phloem of young leaves and stems, affecting photosynthesis and potentially causing the plant to experience stunted growth and/or mortality.[2]
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Disease Progression and Economic Impact (Pennsylvania)

Spotted Lanternflies pose a serious economic threat to multiple U.S. industries, including viticulture, fruit trees, ornamentals and timber.[4] In 2017, at a joint hearing of the Pennsylvania Senate and House Rural Affairs Committee, one vineyard at the core of the infestation zone reported 90% yield loss and corresponding loss in fruit quality.[8] However, some of this loss was due to winter temperatures. In 2018, several other vineyards heavily infested with SLF also reported morality of vines and yield reduction.[8]

V. Management and Control Strategies

Spotted lanternfly quarantine zones have been established across several Northeastern and Mid-Atlantic states to limit the spread of infested materials such as plants, firewood, and outdoor equipment. Pennsylvania, New Jersey, and Ohio have the most extensive restrictions, with statewide quarantines in New Jersey and Ohio and broad county coverage in Pennsylvania. Regulated items include plant material, nursery stock, and equipment, and permits are required for businesses and agencies moving materials within or from quarantined areas. Additional states, including New York, enforce external quarantines requiring inspection of incoming goods from infested regions.

Integrated Pest Management

Integrated pest management strategies for this species are currently limited, with control efforts in the United States focused primarily on early detection, mechanical removal, and public participation. Common methods include scraping and destroying egg masses, using sticky traps to capture nymphs and adults, and applying essential oils as repellents. Chemical controls are used sparingly, as IPM programs prioritize minimizing impacts to non-target species and reducing environmental harm.

Monitoring and Early Detection Methods

Management efforts remain focused on early detection to prevent the continued spread of spotted lanternflies in new locations. Eggs can be removed by hand scraping the masses off the trees. Sticky traps can also be made by wrapping paper around the trees and coating them with adhesive.[2] Certain essential oils, including lavender and linalool, have shown potential as repellents for the spotted lanternfly. Although insecticides may be considered for control, their effectiveness is limited by the insect's high mobility and the difficulty of targeting applications without impacting beneficial insect species [2].

Chemical Control Approaches

Chemical control is not a preferred management strategy in the United States, as IPM emphasizes minimizing pesticide use, accounting for environmental risks, and considering the biology of the insect. In contrast, chemical control remains a primary tool for managing spotted lanternfly populations in China [11]. However, reliance on this approach can be costly, labor-intensive, and may pose risks to human health.

Biological Control

Natural enemies have been shown to contribute to mortality of spotted lanternfly populations. In China, the native egg parasitoid *Anastatus orientalis* has been reported to attack up to 69% of eggs [2], while parasitism rates of approximately 40.2% have been observed in South Korea, where the species has been introduced as a biological control agent [11]. Additionally, the nymphal parasitoid *Dryinus sinicus* has been documented with parasitism rates of up to 30% [9]. Despite these findings, such parasitoids are not currently utilized for biological control in North America, as they are non-native and their potential ecological impacts have not yet been fully evaluated [2].

Cultural Control Practices

Cultural control strategies for the spotted lanternfly include mechanical removal of egg masses, which is recommended for small infestations or residential areas as a nonchemical management approach [9]. However, this method is not practical for large-scale control, particularly because SLFs are attracted to tall structures. This behavior has led to the development of targeted "trap-and-kill" devices, sometimes used in combination with insecticides along habitat edges [9], though their overall effectiveness in reducing populations and plant damage remains uncertain. Removal of the primary host, tree-of-heaven (*Ailanthus altissima*), has also been explored, but it is unclear whether this reduces SLF populations or simply shifts them to nearby hosts. Exclusion netting has shown promise in vineyards, with reductions of up to 99.8% reported [9]; however, further research is needed to determine how these cultural practices can be applied effectively at broader scales.**Regulatory Frameworks and Quarantine Programs**

Federal and state quarantine programs have implemented strict regulatory measures to slow the spread of SLF. The U.S. Department of Agriculture (USDA) and State Departments of Agricultural enforce quarantines in infested regions. These departments require permits for the transporting of materials within infested zones.

VI. Available Datasets and Resources

Image Datasets for SLF and Disease Detection

- Collection of images can be obtained from USDA APHIS and university extension programs with contained labeled of SLF life stages for identification.

VII. Ongoing Research and Future Prospects

Current research is focused on developing sustainable, long-term control strategies for the SLF. Efforts include improving existing tools such as traps, exploring RNA interference (RNAi)-based pest management, and evaluating the role of native U.S. predators. Emerging technologies, including drone-based imaging and machine learning, are also being tested to enhance early detection. Additionally, advancing our understanding of SLF–host plant interactions will be critical for predicting ecological impacts and guiding the development of effective, sustainable management approaches.

VIII. Conclusion

The SLF remains a significant threat to agriculture as it continues to spread, posing both economic and ecological risks in the United States. Complete eradication and highly effective long-term suppression are unlikely due to its broad host range and strong adaptability. As a result, current management strategies emphasize public awareness, continued biological research, and regulatory measures aimed at slowing its spread. Ongoing collaboration among scientists, agencies, and local communities is essential to reduce damage and protect vital agricultural resources.

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